Transportation Capital and City Structure

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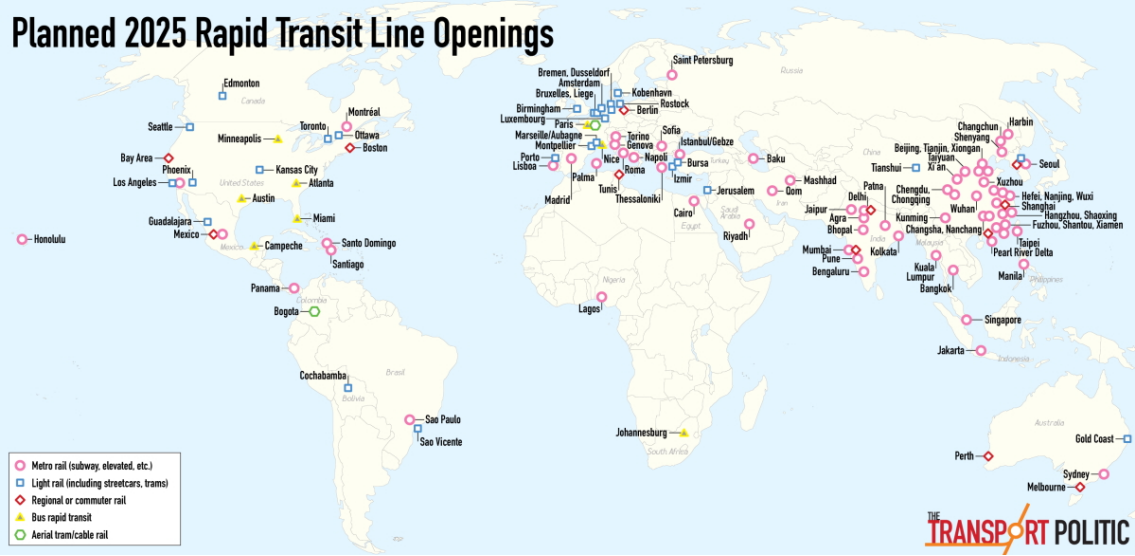
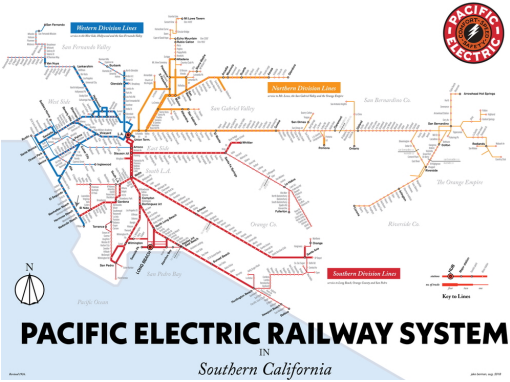
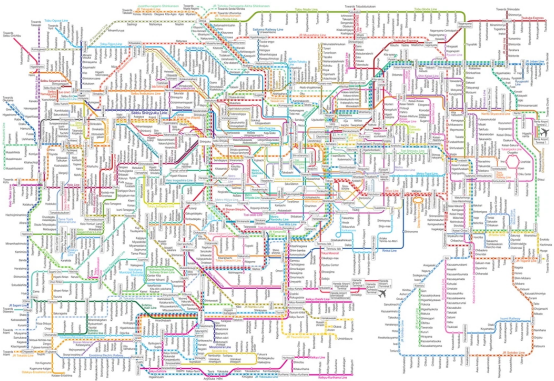
1. What will tomorrow's cities run on?
2. Facts: Correlation between transportation and city shape
3. Framework
4. Empirical Application: U.S. cities?

A lot of public transit efforts!

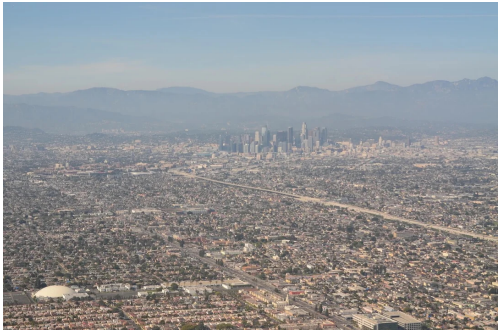


A lot of public transit efforts – Will it last?





City structure – Sprawl for cars and dense for trains



Future cities' structure and transportation?

- Sprawl and road capital
  - Sprawl is more advantageous for cars
  - More cars incentivize building out
- Dense and transit capital
  - Density is more advantageous for trains
  - More trains incentivize building up

$\rightarrow$ Model this feedback loop

Counterfactual – Play back U.S. cities

- Capital: What if car/train investments were less/more?
- Mode: What if they had BRT (Bus Rapid Transit)?
- Technology: What if they had cars with low emissions (EV)?
- Urban: What if building up/out costs were lower/higher?
- Policy: What if land zoning or height regulations changed?

$\rightarrow$ How can we achieve a desired transportation mix?

Literature – One cannot go with the other

- City shapes
  Shape (Harari, 2020) Height (Ahlfeldt et al., 2023) Land use (Coeurdacier et al., 2022)

  Expansion (Rosenthal-Kay, 2024)

- Transportation modes
  Theory (Anas & Moses, 1979) Mode choice (Tsivanidis, 2023)
  Rail suburbanization (Heblich et al., 2020) agglomeration (You, 2021)

  Highway suburbanization (Baum-Snow, 2007) (Redding, 2022)

- Environmental gains from transportation mix

  Subway (Gendron-Carrier et al., 2022), non-econ papers

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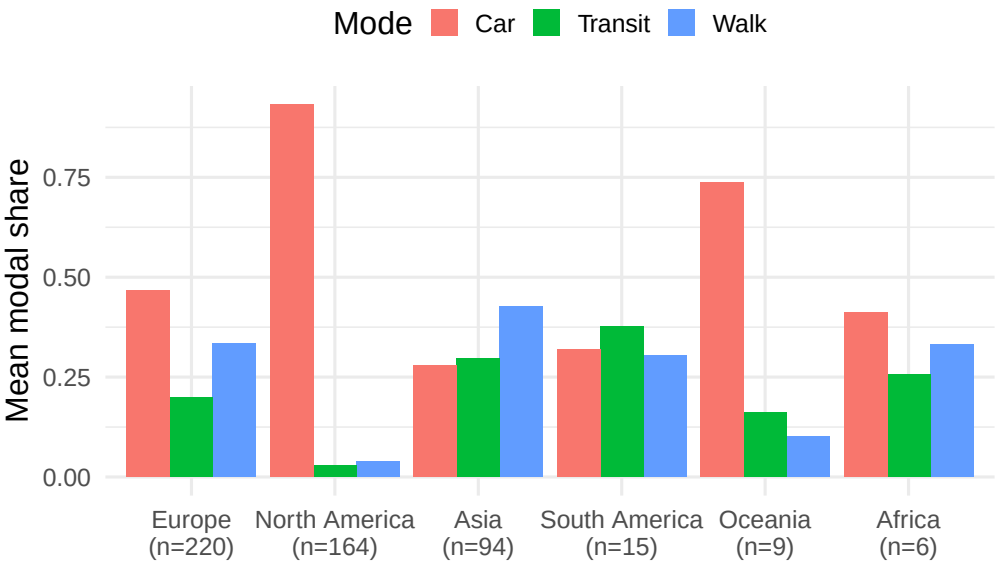
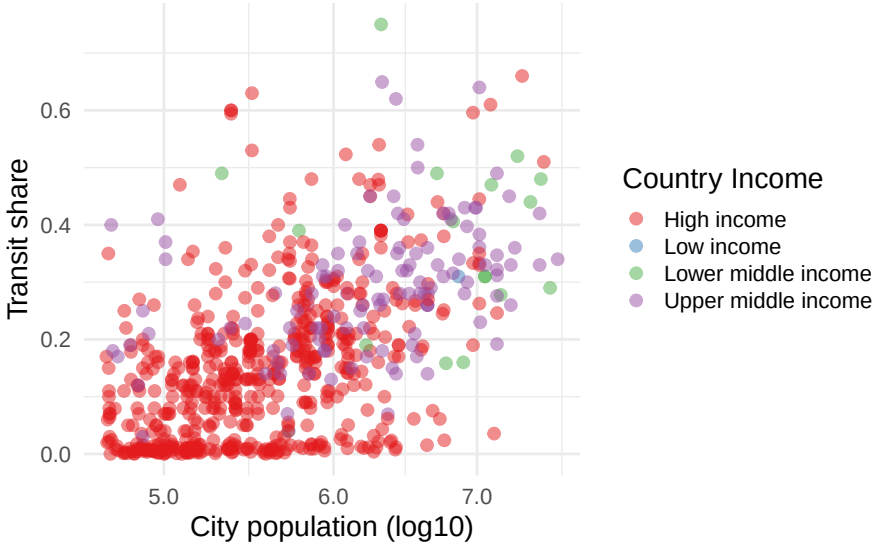
Transportation in cities worldwide (Prieto-Curiel & Ospina, 2024)



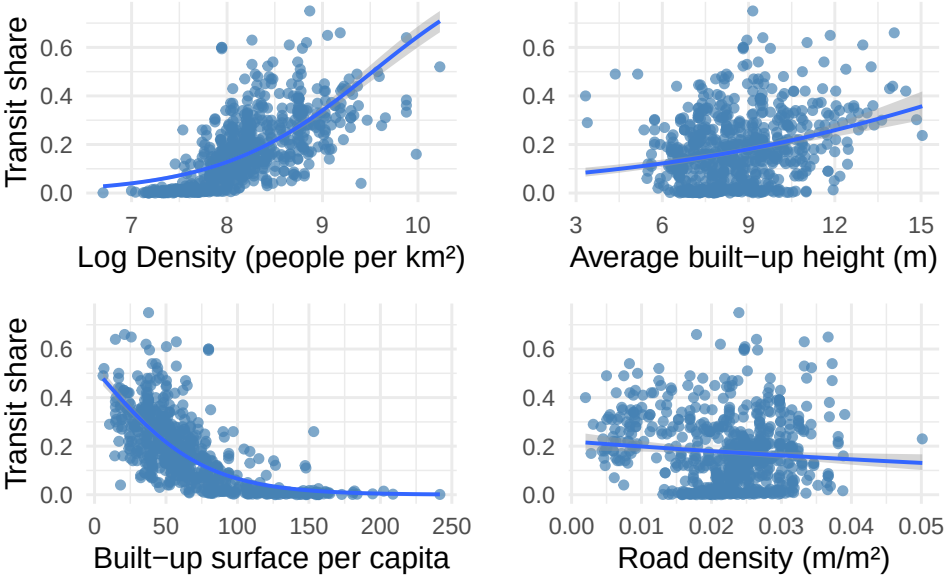
Transit used in less rich and populous cities



Cities database GHSL Stats in the city – A panel (1970-2025) for cities

Variable	Mean	Unit
Urban Centre Area	336.34	km^2
Urban Centre Population	1,863,801.53	people
Total Built-up surface	77,654,714.57	m^2
Average Built-up height	8.64	m
Road length	6,508,933.43	m
Road density	0.02	m/m^2
CO emissions (transport)	1,025,248.65	tons
GHG emissions (transport)	1,042,733.70	tons
NOx emissions (transport)	4,457.26	tons
PM2.5 emissions (transport)	252.91	tons

Correlations – Transit used in dense but not sprawl



Correlation controlling for country income

Dependent Variable:	Transit %			
Model:	(1)	(2)	(3)	(4)
Variables				
Built-up surface per capita	-0.145***			
	(0.024)			
Log Population	0.024***	0.042***	0.020***	0.042**
	(0.001)	(0.007)	(0.003)	(0.008)
Average built-up height		0.050***		
		(0.004)		
Population density			0.152***	
			(0.021)	
Road network density				6.115***
				(0.379)
Fixed-effects				
WB Income Group	Yes	Yes	Yes	Yes
Fit statistics				
R^2	0.537	0.359	0.513	0.411
Observations	705	705	705	705

Clustered (WB Income Group) standard-errors in parentheses

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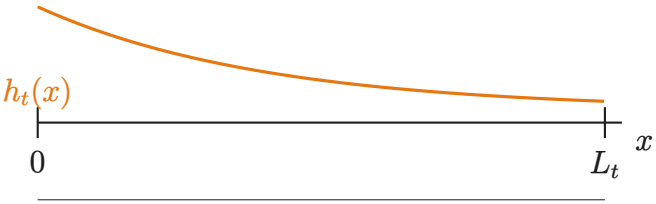
Model the feedback loop

- Dense (built-up height) $\leftrightarrow$ Public Transit
- Sprawl (Built-up surface) $\leftrightarrow$ Cars
- Model the interactions among
  - Households' residential choices and mode share
  - Urban planning (building out vs. up)
  - Transportation investment (government)

Country with a linear city driving output



K_{ct} (Car capital) and K_{rt} (Rail capital)

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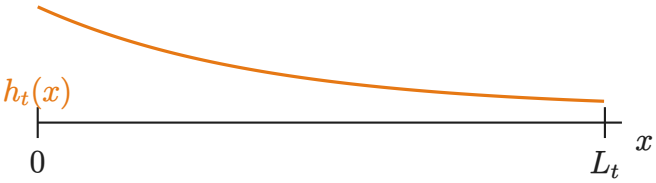

K_{ct} (Car capital) and K_{rt} (Rail capital)

- Homogeneous workers M
- City (service)
  - Population $N_t \equiv \int_0^{L_t} h_t(x)dx$
  - Density $D_t = N_t/L_t$
  - Output $Y_t(N) = N_t^{\gamma+1}$
- Outside (agriculture)
  - Population $M - N_t$
  - Output $(M - N_t)$

How to commute in the city

- $m \in \{car, rail, bus, ?\}$ with cost $c_m(\{K_{m't}\}_{m'}, D_t, L_t; x)$
  - Cheaper with more capital ... $\frac{\partial c_m}{\partial K_m} < 0$
  - **Modes are biased by city shape** ... $\frac{\partial c_{car}}{\partial D_t} > \frac{\partial c_{rail}}{\partial D_t}$, $\frac{\partial c_{car}}{\partial L_t} < \frac{\partial c_{rail}}{\partial L_t}$
    - Cars hate density more than trains (congestion)
    - Trains hate length more than cars (waiting time)
  - Some capital are complementary ... $\frac{\partial c_{bus}}{\partial K_{car} \partial K_{bus}} < 0$

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- From each location x
  - Share of mode m, $s_{jt}(x)$, given by logit share
  - Expected commuting disutility

$$c_t(x) = -\log \sum_m \exp\{-c_{m'}(K_{mt}, D_t, x)\}$$

Floorspace rents clear the spatial equilibrium

- Individual utility living in x

$$u_t(x) = w(N_t) - r_t(x) - c_t(x) = u_{out} \text{ (Spatial Equilibrium)}$$
$$r_t(x) = w(N_t) - c_t(x) - u_{out}$$

- Higher K_{mt} leads to improved transport (c_{mt})
  1. Cheaper expected commute $c(.)$
  2. Higher rents $r(.)$
  3. **City changes** $L_{t+1}, h_{t+1}(x)$
  4. Increase N_t and $w(.)$

Build up $h_t(x)$ and out L_t

- Housing developer solves, for each location x,

$$h_t^*(x) = \arg\max_{h_t(x)} r_t(x)h_t(x) - C(h_t(x))$$

$$h_{t+1}(x) = (1-\delta_h)h_t(x) + \gamma_h(h_t^*(x) - h_t(x))$$

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- The profit at each x, $\Pi(x) \equiv r(x)h^*(x) - C(h(x))$

$$\max_L \int_0^L (\Pi(x) - \Pi_{out})dx$$
$$L_{t+1} = L_t + \gamma_L(\Pi_t(L_t) - \Pi_{out})$$

- γ_h, γ_L represents convex adjustment cost

Transportation investment (government)

- Social welfare for government at time t

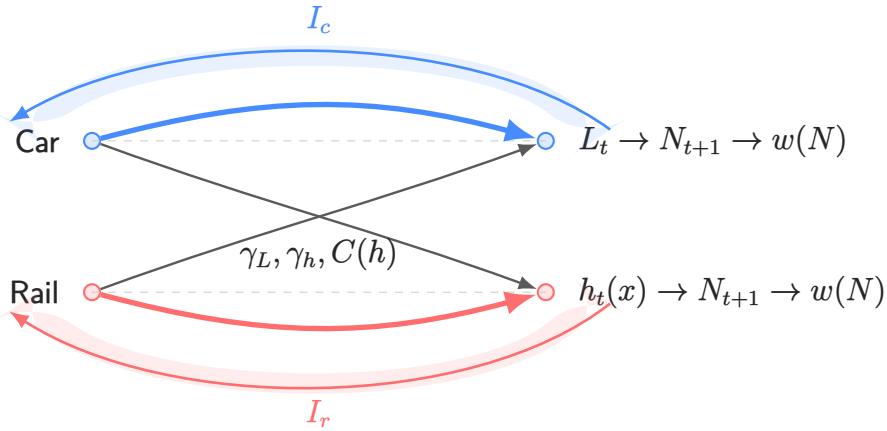
$$W_t = \underbrace{Y_t(N_t) + (M - N_t)}_{\text{Output}} - \underbrace{\int_0^{L_t} h_t(x)c_t(x)dx}_{\text{Commuting Cost}} - \underbrace{\sum_m C_m(I_{mt}, K_{mt})}_{\text{Transport Investment Cost}}$$

- Given the state $S_t = (L_t, \{h_t(x)\}x, \{K_{mt}\}_m)$

$$V(S_t) = \max_{\{I_{mt}\}_m} W_t + \beta V(S_{t+1}|\{I_{mt}\})$$

$$K_{mt_{+1}} = (1 - \delta_m)K_{mt} + I_{mt}$$

Feedback in the model



- Adjustment costs to city shape affect equilibrium selection
  - If building up is impossible ($\gamma_h = 0$), building out wins and then more cars

Additions to the model

- Add emissions per mode to social welfare
  - Governments internalize environmental externalities
  - Technology of modes changes (EV)
- Car adoption involves a fixed cost of use, $F_c/w(N)$
  - Richer countries have potential to have more cars
- Bus rapid transit (BRT) as a distinct mode
  - BRT is complementary to road capital K_{car}
  - May encourage more road investment and car adoption
- Land use of cars and trains (Shoup, 2011)
  - Roads use substantial land area in dense cities (except Musk's Boring)
  - Reduces the efficiency of cars
- ...

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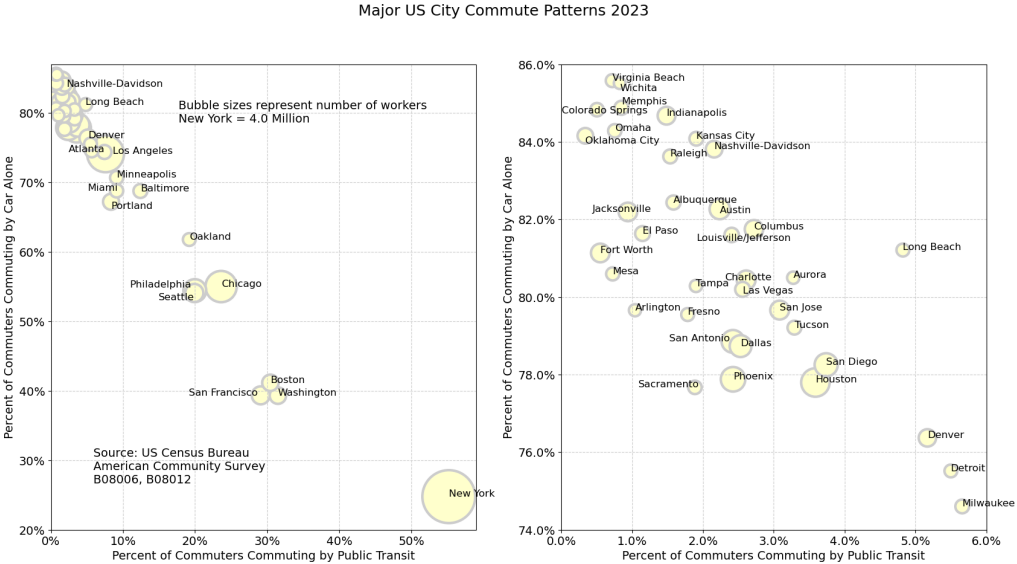
U.S. cities – Transit vs car equilibrium



U.S. cities – Changes also differ by cities

More Than Any Other Large Region, Seattle Has Attracted a Growing Share of Transit Users over the Past 50 Years

Metro areas with the most and least percentage point change in share of commuters using transit from 1970 to 2015–19

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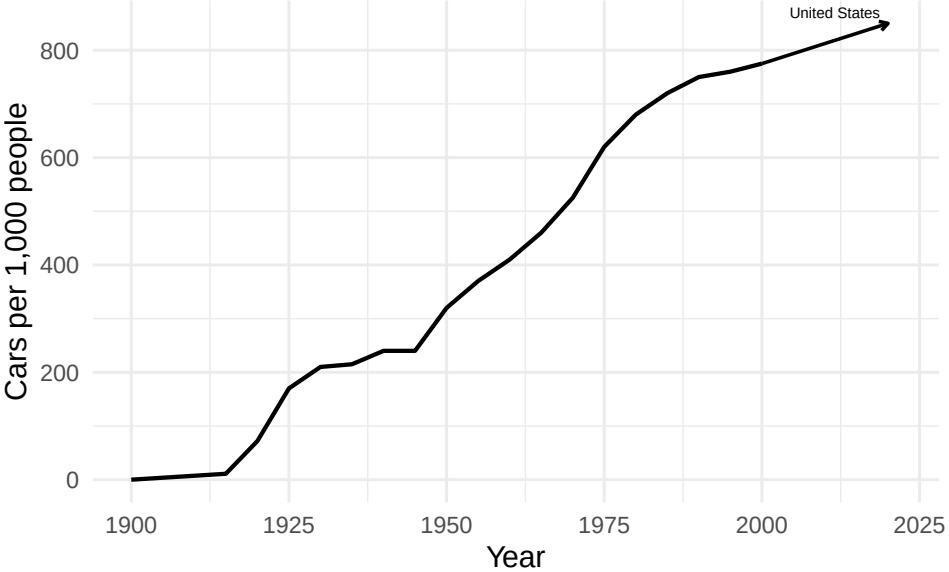
U.S. – Mirror automobile adoption today (Shoup, 2011)



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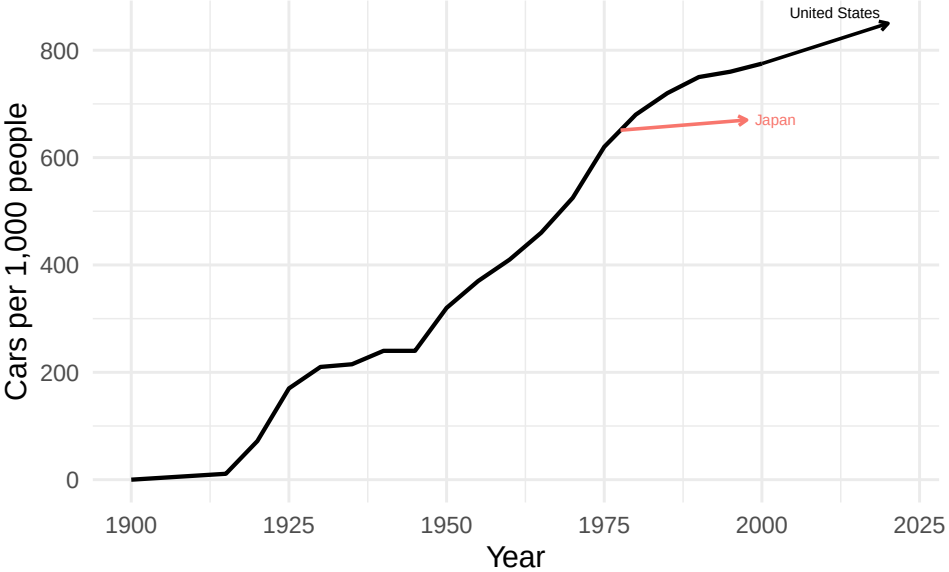

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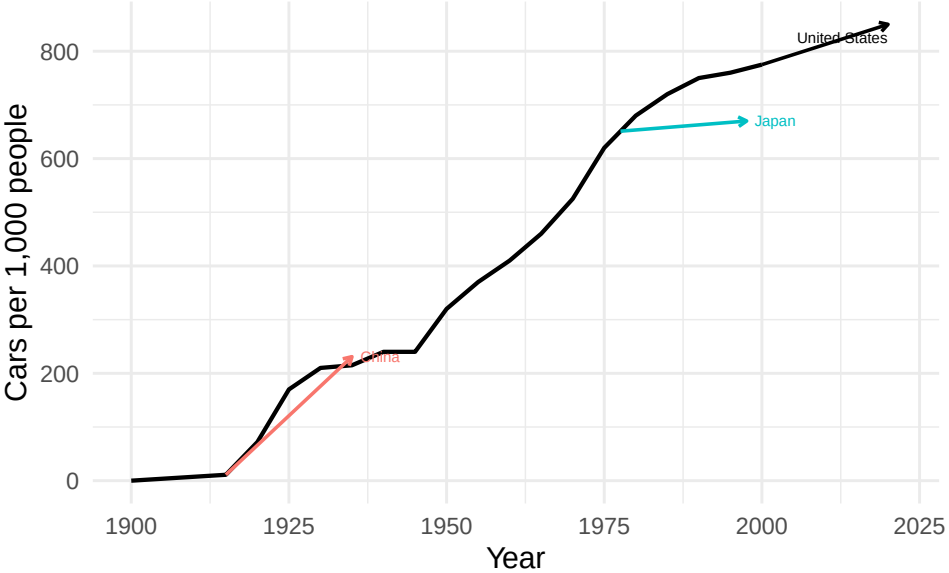

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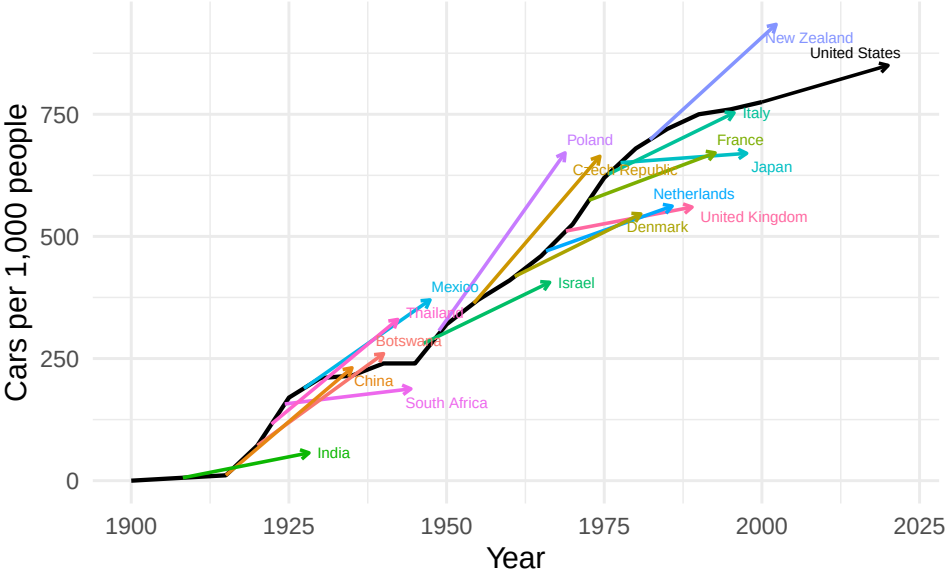

Next steps – Two paths for the empirics

- Cities today globally
  - Data such as built-up land and population are easier to obtain (maybe)
  - May not have had the time for equilibrium selection (developing cities)
- Historical U.S. cities
  - Equilibrium selection probably happened already
  - Many large tug and pull policies
  - Data can be equally granular but require digitization